MIL-M-27733B(USAF)
30 November 1982
SUPERSEDING
MIL-P-27733A(USAF)
23 April 1970

MILITARY SPECIFICATION

# MODIFICATION AND MARKING PROCEDURES FOR TEST EQUIPMENT IN AEROSPACE VEHICLES (CLASS II MODIFICATION)

This specification is approved for use by the Department of the Air Force, and is available for use by all, Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification prescribes procedures for performing Class II modifications (see 6.4.1) on aerospace vehicles, including piloted aircraft and drone aircraft, required to support, research, development, test, and evaluation programs.

## 2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Speifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-W-5088 | Wiring, Aerospace Vehicle |
| MIL-M-43719 | Marking Materials and Markers, Adhesive, Elastomeric, Pigmented, Genera1 Specification For |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |
| MIL-STD-1247 | Marking, Functions and Hazard Designations of Hose, Pipe, and Tube Lines For Aircraft, Missile, and Space Systems |

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:49Z
Check the source to verify that this is the current version before use.

## 3. REQUIREMENTS

### 3.1 Modification data

3.1.1 Class II modification documentation (basic modification installation end demodification) shall be prepared and submitted to the contracting activity in accordance with the Contract Data Requirements List (DD Form 1423) (see 6.2.2).

3.1.2 A final certificate of airworthiness shalll be prepared and placed in the aircraft binder, or flight restrictions imposed as a result of the modification(s) (see 6.4.2).

3.1.3 Flight restrictions shall also be made in the partial flight manual as required by the Contract Data Requirements List (see 6.2.2).

3.1.4 Significant. data concerning the modification shall be recorded on the Significant Historical Data Card (AFTO Form 95). Each modification/demdification accomplished shall be recorded for the lifetime of, and transferred with, the aerospace vehicle.

### 3.2 Modification procedures

3.2.1 Master power switch. A master power switch accessible to the pilot sha11 be installed to do-energize at one time all test equipment and instrumentation installed in conjunction with a Class II modification, except for that required for safe operation of the aircraft or service drone. The switch shall be identified by a plate, finished in black, which has orange borders and orange-filled lettering. It shall be engraved as follows: “CLASS II MODIFICATION EQUIPMENT MASTER POWER SWITCH".

3.2.2 Emergency egress. - Modified components and equipment installed in conjunction with Class II modifications shall not in any manner compromise crew safety or escape from the aircraft in flight or on the ground. In cases where modifications were made in aircraft using power assisted devices in the crew escape system, particular care shall be taken to assure that propellant actuated systems, safety devices, and components are not rendered inaccessible for inspection. No heat-generating or heat-emitting equipment shall be installed in proximity of any propellant-actuated device.

3.2.3 Cutting and drilling. When the modification results in cutting, drilling, or disturbing any portion of the fuselage or structure that forms the pressure vessel for cabin pressurization, the area shall be resealed as specified by the modification engineering authority (see 6.4.2). If cuts or structural repairs are made in the pressure vessel, aircraft pressurization shall be checked as specified in the contract (see 6.2.lb).

3.2.4 Wiring. Electrical wiring shall be selected for use in accordance with MIL-W-5088. Cable or wire with polyvinyl chloride (PVC) insulation shall not be used in manned aerospace application, in accordance with MIL-STD-454.



Source: https://assist.dla.mil -- Downloaded: 202610-20T12:49Z
Check the source to verify that this is the current version before use.

3.3 Idenftification and marking

3.3.1 Advisory plate. An advisory plate shall be installed in a conspicuous (visible) location in the cockpit. of manned aircraft. This plate shall be finished in black, which has orange borders and white lettering. It shall read as follows, "CLASS II MODIFICATION EQUIPMENT INSTALLED--SEE OPERATING OR MAINTENANCE INSTRUCTIONS".

3.3.2 Switches, power circuit breaker protective devices, and controls used in the modification shall be identified by a plate which surrounds and incorporates the switch, device or control. The plate shall be finished in black, which has orange borders, and engraved with white-filled lettering and numbers (see figure 1). This method shall also be used to identify a related group of equipments (see figure 2). When a durable identification plate cannot be installed because of q aterial or space limitations, the identification shall be accomplished by application of a solid or broken line of orange paint around the equipment involved.

3.3.3 Test equipment carriers.

3.3.3.1 Fuel tanks. When fuel tanks are used as carriers for test equipment, they shall be identified by an encircling four-inch band of orange located approximately one-quarter of the tank length, from the leading edge. All filler caps shall be secured and all single-point fuel and air lines to the affected tank(s) shall be rendered inoperative to prevent inadvertent fueling. The inscription "CLASS II MODIFICATION EQUIPMENT INSTALLED-DO NOT FUEL" shall be stenciled in orange adjacent to the filler cap(s). The modification number (see 6.4.3) shall be stenciled in orange, near the base.

3.3.3.2 Non-standard tanks and stores. Non-standard external tanks and stores used as carriers for test equipment shall be marked with a four-inch orange band around the tank or store, located approximately one-quarter of the tank or store length, from the leading edge. The Base Supply Account Number of the originating installation for the tank or store, and the aircraft tail number, shall be stenciled on both sides, and as near the center as possible, using one-inch numbers. Tanks (but not stores) shall have the modification number stenciled near the base, with one-half inch orange lettering and numbers.

3.3.4 Equipment. Group A and B (see 6.4.4 and 6.4.5) q edified equipment, brackets and racks, shall be marked with sufficient orange coloring to permit ready identification of the modified portions of the Class II modification project, and to facilitate restoration to standard configuration.

3.3.5 Components

3.3.5.1 Group A and B external components (such as antennas, refueling booms, probes, and radomes) shall be stenciled with the modification number near the base, with one-half inch orange lettering and numbers.

3.3.5.2 Group A and B components (such as cameras, inverters, junction boxes, amplifiers, control boxes, recorders, and test equipment) shall be identified by a modification label. The label shall be orange in color and appear as illustrated on figure 3 or figure 4.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:49Z
Check the source to verify that this is the current version before use.

3.3.5.2.1 Where space Is limited, the words "ITEMS", “MOD #“, etc., may be left off the label and just the information entered. The shortened label shall be orange in color and appear as illustrated on figure 4.

3.3.5.2.2 The identification label shall be so positioned that it is readily visible when installed. If it cannot be positioned where it is readily visible, it may be located in a less visible position, provided that the face of the equipment is marked with orange or is q ounted in a panel containing only modified components.

3.3.5.3 Small modified components (such as altimeter, temperature measuring or special pressure gauges) shall be painted orange when there is a possibility of confusion with unmodified items. For example, a tachometer-generator mounted on an engine shall be painted orange to make it readily identifiable during an engine change. A modification identification label would not be required because it would not survive in the engine environment.

3.3.6 Specially calibrated instruments. Specially colibrated instruments, installed as part of the crew station instrumentation or in photo or observer’s panels, shall be identified by an orange dot, approximately one-fourth of an inch in diameter, on the face of the instrument, so located as not to interfere with the pointer or dial markings.

3.3.7 Electrical wiring. Electrical wiring for Class II modified aerospace vehicles and components shall be identified by numerals and letters marked on the wiring in accordance with MIL-W-5088. Identification numberals and letters on wiring used in the modification shall be orange in color. Alternate use of electrical wiring having orange-colored outer covering and marked with black numberals and lettering, is permitted. Wire inside an enclosed component (including instrumentation) not having a standard number already assigned, in accordance with MIL-U-5088, shall show a number prefixed by letters (see figure 5) furnished by the modification engineering authority for the Class II modification project (see 6.2.1.c). A record of the assigned numbers shall be maintained by the modification engineering authority for the project to prevent duplication.

3.3.7.1 Coaxial, thermocouple, twisted shielded, prefabricated cable, and similar types of cable that cannot be marked in this manner by wire-marking machines shall be identified by orange numerals and lettering, marked in accordance with MIL-W-5088. on sections of black plastic tubing or tape installed over the cable. Wire either too small, or because of dielectric (e.g., some of the Teflons) shall be identified by this method. Prefabricated and multiple-conductor cables need not have each wire identified at each end, but they must be completely identified on the wiring diagrams maintained by the modification engineering authority for the project. Prefabricated and multiple-conductor cables shall also be identified with an orange band at not more than six-foot intervals, or as specified by the q edification engineering authority for the project.

3.3.7.2 when electrical connectors of the same size and type are used in close proximity, the plug shells shall be color-coded to their receptacles to prevent mismatching.

3.3.8 Tubing shall be color-coded in accordance with MIL-STD-1247, with one band of orange tape adjacent to and following the standard color code. The tape shall be placed as illustrated in figure 6.



Source: https://assist.dla.mil -- Downloaded: 2026l0-20T12:49Z
Check the source to verify that this is the currentversion before use.

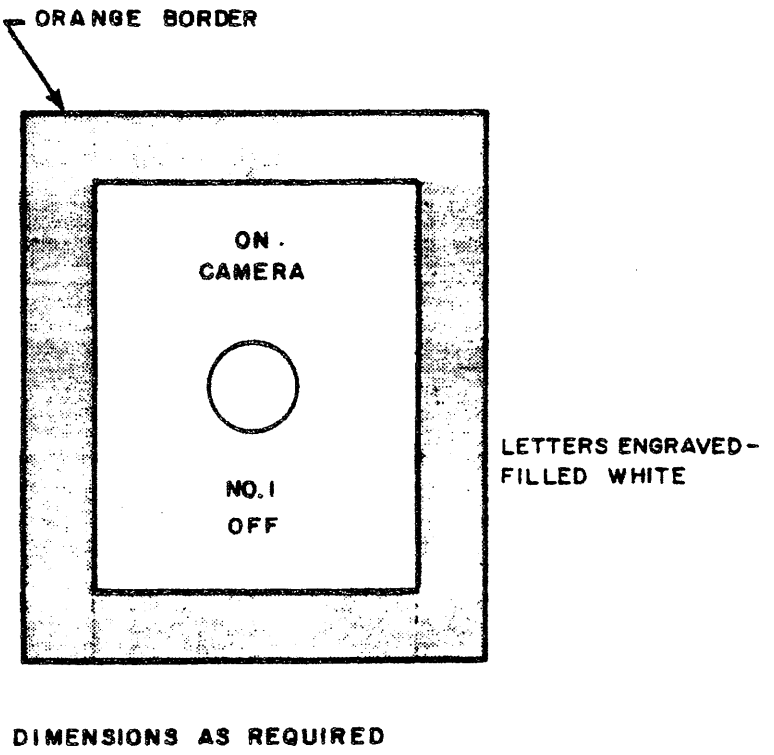



FIGURE 1. Sample of tropical specific control identification plate.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:49Z
Check the source to verify that this is the current version before use.

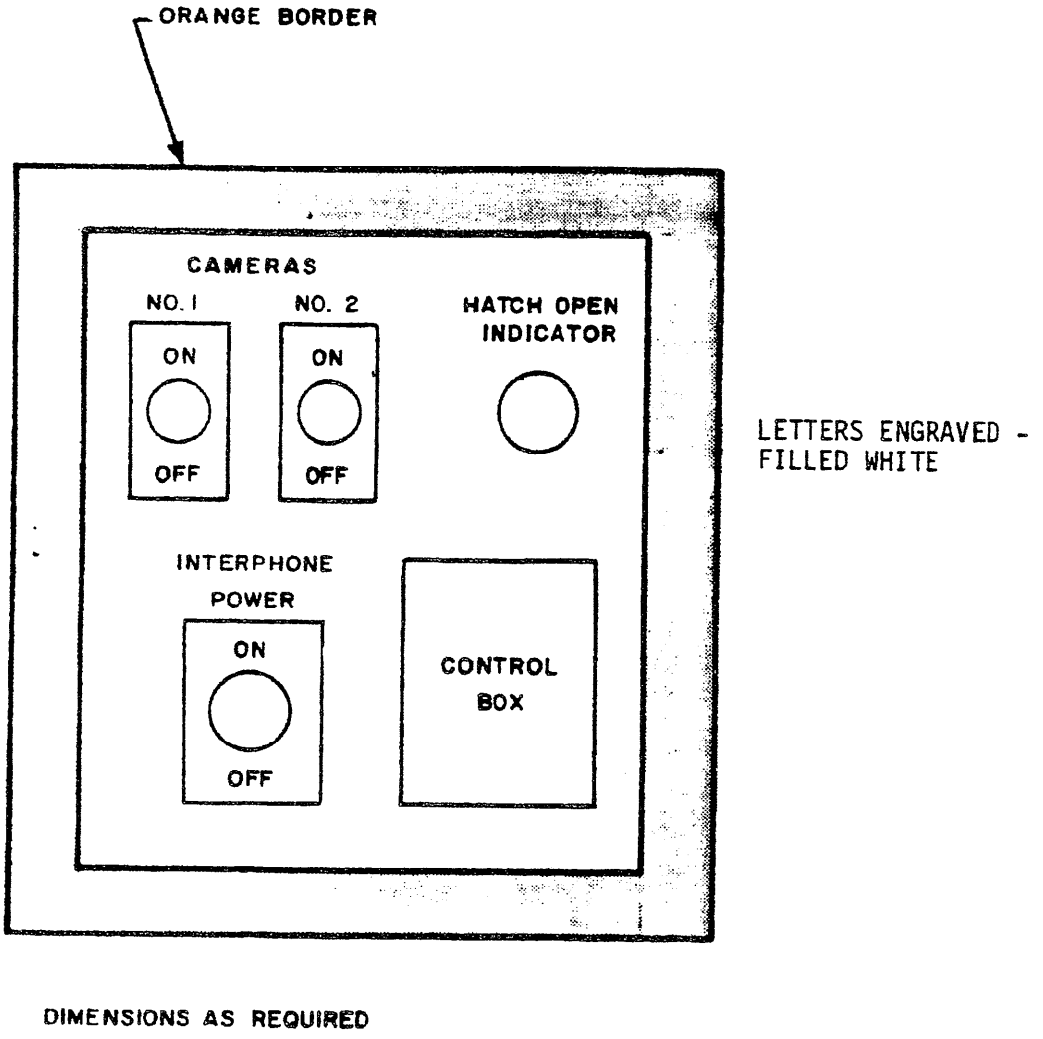
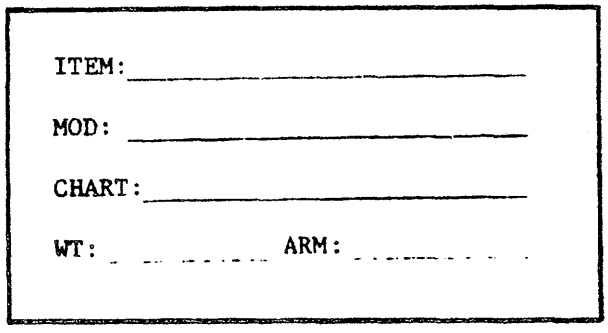



FIGURE 2. Sample of typical general controls identification plate.



Source: https://assist.dla.mil -- Downloaded: 202610-20T12:49Z
Check the source to verify that this is the current version before use.




ITEM----------shall be the nomenclature of the component, abbreviated as necessary to fit the limited space available.
Mod #-------shall be the modification number (if available) or the project title.
CHART “A” #--shall be the number used to identify the compo nent in the aircraft weight and balance, Chart A (DD Form 365A)
WT----------shall be the item weight
ARM---------shall be the fuselage station in inches.

FIGURE 6. Coponents identification label.

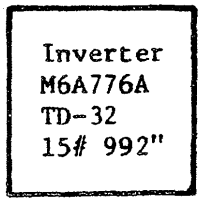
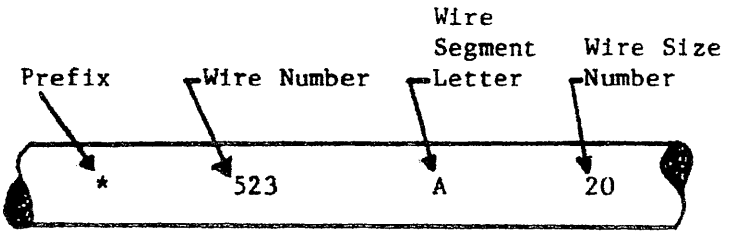



FIGURE 4. Components identification limted space.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:49Z
Check the source to verify that this is the current version before use.




*Letter prefix to be furnished by the modification engineering authority for the Class II modification project.

FIGURE 5. Sample fo a typical wire number assignment

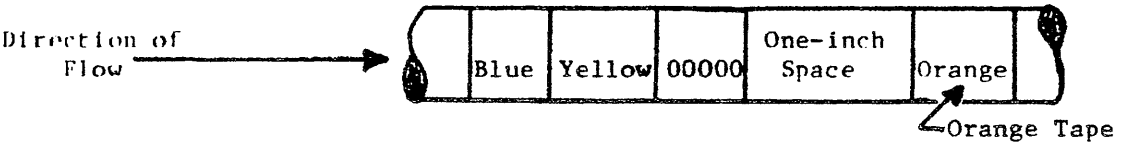



FIGURE 6. Marking of typical tube in a hydraulic system with te standard color code tape



Source: https://assist.dla.mil -- Downloaded: 202610-20T12:49Z
Check the source to verify that this is the current version before use.

3.4 Technical acceptance. Prior to acceptance of the Class II modification, the aerospace vehicle shall be checked as specified in section 4 herein, to demonstrate proper operational qualities (see 6.2.1.d).

3.5 Workmanship. All details of workmanship shall be in accordance with high-grade installation practice. Insofar as practicable, the modification shall be so accomplished that the aerospace vehicle can be readily returned to standard configuration, except when an aircraft has been redesignated with an "N" classification prefix, or when drones or target missiles will be tested to destruction.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Examination of product. Each modification shall be thoroughly examined to determine compliance with requirements herein. A physical configuration inspection (see 6.4.6) and equipment inventory shall be made to determine the accuracy of identification and documentation, and adequacy of maintenance and operation instructions as specified in the Contract Data Requirements List (see 6.2.2).

4.3 The following checks and evaluations shall be accomplished for all manned aircraft.

4.3.1 Operational ground check. To assure that the aerospace vehicle is safe for flight, all systems and components affected by the modification shall be operationally ground checked to determine safe operation and condition prior to release for the project test flight. Systems which cannot be adequately ground checked may be flight checked provided that flight safety is not degraded and provided that a power-on check has been accomplished.

4.3.2 Aerodynamic evaluation flight. This may be flown in conjunction with a functional check flight (FCF). It is an evaluation of aircraft structure or aerodynamic characteristics, and is performed by flight test pilots to determine the effects of the modification on the aircraft. It is required whenever the modification changes aerodynamic configuration of the aircraft, changes Flight controls, or affects structural strength. A minimum crew, consistent with the flight mission requirements will be aboard.

4.3.3 Functional check flight. If determined by the maintenance officer of the unit to which the aircraft, is assigned, a functional check flight or the aircraft shall be performed to verify airworthiness of the aircraft and associated systems or components that have been modified or affected by modification as specified in the contract {see 6.2.1.b and b.2.1.d).



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:49Z
Check the source to verify that this is the current version before use.

## 5. PACKING

5.1 This section is not applicable to this specification.

## 6. NOTES

6.1 Intended use. The procedures covered by this specification are intended for use in accomplishing Class II modifications, are intended to require Important desirable characteristics, and to prohibit serious installation design discrepancies. This specification cannot properly account for special problems encountered in the installation of new equipment. Contractors are encouraged to continually review the requirements of this specification and supporting specifications, with the intent of developing waiver proposals when such waivers will yield improved operations while providing adequate margins of safety.

6.2 Ordering data

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. The following Technical Order(s) are applicable to the requirements of this specification and should be called out In the contract by reference(s) or detailed requirements.

| Paragraph No. | Requirement Title | Applicable T.O. | Option |
|---|---|---|---|
| 3.2.3 | Aircraft pressuration checked and tested | -2 Series T.O.'s Maintenance Instructions" | ------ |
| | | -3 Series T.O.'s “Structural Repairs” | ------ |
| 4.4 | Verification of airworthiness | T.O. 1-1-300 "Acceptance/Functional Check Flight and Maintenance Operator Checks" | ------ |
| | | 4 Series T.O.'s "Inspection Requirements" | ------ |

c. The type of prefix to be used on electrical wiring (see 3.3.7)

d. Whether a functionally check flight (FCF) is required (see 3.4 and 4.3.2).



Source: https://assist.dla.mil -- Downloaded: 202610-20T12:49Z
Check the source to verify that this is the current version before use.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

| Paragraph No. | Data Requirements Title | Applicable DID No. | Option |
|---|---|---|---|
| 3.1.1 | Modification package | DI-E-3115 | - - - - |
| 3.1.2 | Partial flight manual | DI-E-3115 | - - - - |
| 4.2 | Operating and maintenanee instructions | DI-E-3115 | - - - - |

(Copies of data item descriptions required in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120, or as directed by the Contracting officer.)

6.3 The following Technical Orders may be pertinent to the Class II modification and are listed for information purposes:

GENERAL ENGINEERING MANUALS

| | |
|---|---|
| T.O. 1-1A-1 | Engineering Handbook Series For Aircraft Repairs--General Manual For Structural Repairs |
| T.O. 1-1A-8 | Engineering Manual Series --Aircraft and Missile Repair--Structural Hardware |
| T.O. 1-1AE-9 | Engineering Series For Aircraft Repair-Aerospace Metals--General Data and Usage Factors |
| T.O. 1-1A-12 | Engineering Handbook Series for Aircraft Repairs--Fabrication, Maintenance and Repair of Transparent plastics |
| T.O. 1-1A-14 | Installation Practices For Aircraft Electrical and Electronic Wiring |

(Copies of Technical Orders required in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer, or in accordance with Technical Order 00-5-2.)

6.4 Definitions

6.4.1 Class II modifications. Class II modifications are temporary hardware or software changes or alterations to aerospace vehicles (aircraft, guided weapons, drones, remotely piloted vehicles (RPV), and missiles other than strategic), airborne support equipment, external and internal stores, subsystems, components, or support equipment that interface with an aerospace



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:49Z
Check the source to verify that this is the current version before use.

vehicle. They are installations or changes which support research development, design changes to existing Class II modifications, and development and oerational test and evaluation programs or in-service testing of systems or equipment.

6.4.2 Modification engineering authority. Activity having program management responsibility for an engineering effort required to accomplish integration or installation of Group B items which have been previously developed and tested or which have been commercially used in the same environment.

6.4-3 Modification number. Number issued by the Configuration Management Officer responsible for the Class II modificatio project. Modifications, demodifications, partial modifications, reidentification, or changes shall be identified by a seven-digit alphabetical/numerical designator, as follows:

M-----Modification

2----last digit of calendar year

A-----first basic modification to aircraft for year

399-----last 3 digits of aircraft serial number (aircraft tail number)

A-----change indicator to basic modification (e.g., A=initial, B=first change to basic mod)

Example:

M2A399A - first basic modification in 1982 to aircraft tall number 399

M2A399B - first change to 1982's first basic modification to aircraft tail number 399

M2B399A - second basic modification in 1982 to aircraft tail number 399

M2B399B - first. change to second basic modification

(1) Basic Modification: Installation of equipment to provide a new or added capacity. The letter “O” will not be used in designators. If the number of modifications in a single year exceed 25, the first digit of the aircraft serial number will be dropped and an additional alphabetical designator added; e.g., M2ZA99A.

(2) Change to any Basic Modification: The letter “O” will not be used. If the number of changes exceed 25, the first digit of the aircraft serial number will be dropped and an additional alphabetical designator added at the end; e.g., M2B99ZA.

(3) Demodification: Removal of basic modification. Demodifications will be identified with the original modification number.



Source: https://assist.dla.mil -- Downloaded: 202610-20T12:49Z
Check the source to verify that this is the current version before use.

6.4.4 Group A. The items, parts, or components to be installed in an aerospace vehicle to support, secure, interconnect, or accommodate the Group B components.

6.4.5 Group B. The equipment which, when installed on an aerospace vehicle with Group A items, completes a modification. Normally, Group B items are removable.

6.4.6 Physical configuration inspection (PCI). A detailed inspection of the modification to ensure the modification has been made as prescribed in the modification package.

6.5 Changes from the previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Air Force - 11

Preparing activity:
Air Force - 11

Project No. 15GP-F039



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:49Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:49Z
Check the source to verify that this is the current version before use.

**INSTRUCTIONS:** In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

EPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

**OFFICIAL BUSINESS**
PENALTY FOR PRIVATE USE $300

**BUSINESS REPLY MAIL**
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

ASD/ENESS
Wright-Patterson AFB, OH 45433

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:49Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)* |

5. PROBLEM AREA

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 1426 82 MAR

PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:49Z
Check the source to verify that this is the current version before use.